

ORIGINAL ARTICLE

Abiraterone and Olaparib for Metastatic Castration-Resistant Prostate Cancer

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Abstract

BACKGROUND Preclinical studies and results of a phase 2 trial of abiraterone and olaparib suggest a combined antitumor effect when the poly(adenosine diphosphate[ADP]-ribose) polymerase inhibitor olaparib is combined with next-generation hormonal agent abiraterone to treat metastatic castration-resistant prostate cancer (mCRPC).

METHODS We conducted a double-blind, phase 3 trial of abiraterone and olaparib versus abiraterone and placebo in patients with mCRPC in the first-line setting. Patients were enrolled regardless of homologous recombination repair gene mutation (HRRm) status. HRRm status was determined following enrollment by tumor tissue and circulating tumor DNA tests. Patients were randomly assigned (1:1) to receive abiraterone (1000 mg once daily) plus prednisone or prednisolone with either olaparib (300 mg twice daily) or placebo. The primary end point was imaging-based progression-free survival (ibPFS) by investigator assessment. Overall survival was among the secondary end points.

RESULTS At this planned primary analysis at the first data cutoff, median ibPFS was significantly longer in the abiraterone and olaparib arm than in the abiraterone and placebo arm (24.8 vs. 16.6 months; hazard ratio, 0.66; 95% confidence interval [CI], 0.54 to 0.81; $P < 0.001$) and was consistent with blinded independent central review (hazard ratio, 0.61; 95% CI, 0.49 to 0.74). At this data cutoff, overall survival data were immature (28.6% maturity; hazard ratio, 0.86; 95% CI, 0.66 to 1.12; $P = 0.29$). The safety profile of olaparib and abiraterone was consistent with the known safety profiles of the individual drugs. The most common adverse events in the abiraterone and olaparib arm were anemia, fatigue/asthenia, and nausea.

*A complete list of investigators in the PROpel trial is provided in the Supplementary Appendix, available at evidence.nejm.org.

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CONCLUSIONS At primary analysis at this first data cutoff, abiraterone combined with olaparib significantly prolonged ibPFS compared with abiraterone and placebo as first-line treatment for patients with mCRPC enrolled irrespective of HRRm status. (Funded by AstraZeneca and Merck Sharp & Dohme, LLC, a subsidiary of Merck & Co., Inc., Rahway, NJ, USA; ClinicalTrials.gov number, [NCT03732820](#).)

Introduction

Current treatment options in the first-line setting of metastatic castration-resistant prostate cancer (mCRPC) mainly comprise next-generation hormonal agents (e.g., abiraterone and enzalutamide) or docetaxel chemotherapy.¹ Despite the reported activity of these agents, overall survival in clinical trial settings is approximately 3 years.²⁻⁴ In the real-world setting, overall survival is shorter, and approximately 50% of patients receive only one life-prolonging therapy.⁵

The phase 3 PROfound trial of the poly(adenosine diphosphate[ADP]-ribose) polymerase (PARP) inhibitor olaparib versus enzalutamide or abiraterone demonstrated significant imaging-based progression-free survival (ibPFS) and overall survival benefits with olaparib in patients with mCRPC harboring *BRCA1*, *BRCA2*, and *ATM* mutations whose disease had progressed on a next-generation hormonal agent. Significant ibPFS and a trend toward prolonged overall survival in the overall trial population with alterations in **homologous recombination repair (HRR)** genes was also observed.^{6,7}

Preclinical models suggest that when PARP inhibitors are combined with next-generation hormonal agents, there may be a combined antitumor effect.⁸⁻¹⁰ This is potentially because of PARP involvement in positive coregulation of androgen receptor (AR) signaling, which leads to enhanced AR target gene suppression when PARP/AR signaling is co-inhibited.⁹ Other studies have reported that next-generation hormonal agents inhibit transcription of some HRR genes, thereby inducing HRR deficiency and increased sensitivity to PARP inhibitors through nongenetic mechanisms.^{8,10} These preclinical findings were confirmed in a phase 2 trial that showed that combining olaparib with abiraterone to treat patients with mCRPC who had previously received docetaxel, **and were unselected by HRRm status**, resulted in significantly longer

ibPFS compared with abiraterone and placebo (hazard ratio, 0.65; 95% confidence interval [CI], 0.44 to 0.97; $P=0.034$).¹¹ Prespecified and additional post hoc analyses from the study suggested that a treatment effect was observed in patients with and without HRRm.^{11,12}

PROpel is a double-blind, randomized phase 3 trial of abiraterone and olaparib versus abiraterone and placebo in first-line treatment of patients with mCRPC. The primary end point was efficacy by investigator-assessed ibPFS. We report the planned primary analysis from the first data cutoff of the trial.

Methods

PATIENTS

Eligible patients were 18 years of age or older (or 19 years of age or older in South Korea) and had histologically or cytologically confirmed prostate adenocarcinoma with at least one documented metastatic lesion on a bone scan or computed tomography or magnetic resonance imaging scan. **With the exception of androgen depletion therapy and first-generation antiandrogen agents with a 4-week washout period, prior systemic treatment in the mCRPC first-line setting was not allowed. Docetaxel during neoadjuvant/adjuvant treatment for localized prostate cancer and metastatic hormone-sensitive prostate cancer (mHSPC) was permitted.**^{13,14} Full eligibility criteria can be found in the trial protocol (p. 221) available with the full text of this article at [evidence.nejm.org](#). All patients provided written informed consent.

TRIAL DESIGN AND INTERVENTIONS

This was a double-blind, placebo-controlled phase 3 trial. Eligible patients were randomly assigned (1:1) to abiraterone (1000 mg once daily) in combination with either olaparib (300 mg twice daily) or placebo. All patients received prednisone or prednisolone (5 mg twice daily) per the abiraterone label requirement. Random assignment was stratified by distant metastasis type (bone only, visceral, or other; see definitions on Supplementary Appendix p. 8) at baseline and by docetaxel treatment at the mHSPC stage of disease (yes or no). Study treatment continued until objective imaging-based progressive disease as assessed by the investigator (using Response Evaluation Criteria in Solid Tumors 1.1¹⁵ for soft tissue lesions and Prostate Cancer Working Group-3¹⁶ criteria for bone lesions),

unacceptable toxicity, or withdrawal of consent. Following objective disease progression, further treatment was at investigator discretion. Patients could continue study treatment if the investigator believed, and the AstraZeneca Study Physician agreed, that the patient could continue to receive clinical benefit and was not experiencing serious toxicity and no better alternative treatment was available. Crossover from placebo to receive olaparib in combination with abiraterone was not allowed. Risk mitigation factors for Covid-19 related to study conduct and patient management were implemented (trial protocol, p. 338).

END POINTS

The primary end point was investigator-assessed ibPFS or death from any cause in the absence of disease progression. Sensitivity analysis by blinded independent central review of imaging and exploratory subgroup analysis of investigator-assessed ibPFS were also conducted. Subgroup analysis assessed treatment effect consistency across pre-specified prognostic factors of potential importance, including HRRm status (Supplementary Appendix, p. 8).

Patient enrollment was not based on HRRm status; however, HRRm testing was predefined. Both tumor tissue (mostly archival) and blood samples at baseline were collected from more than 98% of randomly assigned patients so that HRRm status by both tumor tissue test (FoundationOne CDX) and a circulating tumor (ct)DNA-based test (FoundationOne Liquid CDx test) could be determined after randomization and before primary analysis. Although tumor tissue testing is the gold standard for HRRm testing, a high test failure rate in prostate tumor samples (approximately 30%) has been reported in previous studies.^{6,17} Considering the challenges of obtaining suitable tumor tissue from some patients and the inherent performance characteristics of tumor tissue and plasma ctDNA HRRm tests, both tumor tissue and ctDNA-based tests were used. To ensure comprehensive evaluation of HRRm status, post hoc analysis of aggregate tumor tissue and ctDNA data was conducted to maximize the proportion of patients with assigned HRRm status and to minimize potential false negatives. The genes assessed by HRRm testing were *ATM*, *BRCA1*, *BRCA2*, *BARD1*, *BRIP1*, *CDK12*, *CHEK1*, *CHEK2*, *FANCL*, *PALB2*, *RAD51B*, *RAD51C*, *RAD51D*, and *RAD54L*. Patients were classified into three groups: HRRm, non-HRRm, and HRRm unknown. The HRRm group comprised patients with any deleterious or suspected deleterious HRR gene mutation detected. The non-HRRm group comprised patients with no deleterious or suspected deleterious HRR

Table 1. Characteristics of Patients at Baseline.*

Characteristic	Abiraterone and Olaparib (n=399)	Abiraterone and Placebo (n=397)
Age at random assignment — median (range), yr	69.0 (43–91)	70.0 (46–88)
Gleason score		
≥8	265 (66.4)	258 (65.0)
Missing	13 (3.3)	5 (1.3)
ECOG performance status		
0 (normal activity)	286 (71.7)	272 (68.5)
1 (restricted activity)	112 (28.1)	124 (31.2)
Missing	1 (0.3)	1 (0.3)
Prior docetaxel treatment		
Yes	97 (24.3)	98 (24.7)
At mHSPC stage	90 (22.6)	89 (22.4)
Prior treatment with NHA†		
Yes	1 (0.3)	0
Disease site‡		
Bone	349 (87.5)	339 (85.4)
Distant lymph nodes	133 (33.3)	119 (30.0)
Locoregional lymph nodes	82 (20.6)	89 (22.4)
Prostate and adjacent structures	47 (11.8)	46 (11.6)
Respiratory (including lung)	40 (10.0)	42 (10.6)
Liver	15 (3.8)	18 (4.5)
HRRm status (aggregate)§		
HRRm	111 (27.8)	115 (29.0)
Non-HRRm	279 (69.9)	273 (68.8)
HRRm unknown	9 (2.3)	9 (2.3)
BRCa prevalence (aggregate)		
BRCA1	9 (2.3)	3 (0.8)
BRCA2	38 (9.5)	35 (8.8)
Baseline serum PSA — median (IQR), µg per liter	17.90 (6.09–67.00)	16.81 (6.26–53.30)

* Data were derived from electronic case report forms. Values are presented as No. (%) unless specified otherwise. A more detailed table of baseline characteristics can be found in Table S1. BRCa denotes BRCA mutation, ECOG Eastern Cooperative Oncology Group, HRRm homologous recombination repair gene mutation, IQR interquartile range, mHSPC metastatic hormone-sensitive prostate cancer, NHA next-generation hormonal agent, and serum-PSA Serum Prostate-Specific Antigen.

† Before metastatic castration-resistant prostate cancer, treatment with next-generation hormonal agents (except abiraterone) was allowed, provided patients had not had PSA or clinical or imaging-based progression during the treatment and the treatment was stopped at least 12 months before random assignment.

‡ Investigators could select more than one disease site. Entries for “other locally advanced sites,” “other distant sites,” and “other” have been excluded.

§ HRRm comprises patients with any deleterious or suspected deleterious HRR gene mutation detected; non-HRRm comprises patients with no deleterious or suspected deleterious HRR gene mutation detected and HRRm unknown denotes patients for whom mutation testing was not performed, mutation testing failed because of insufficient quantity or quality of sample, or technical failure occurred at sequencing or postsequencing steps on analysis.

gene mutation detected. Finally, the HRRm unknown group comprised patients for whom mutation testing was not performed because of a lack of or inadequate samples, mutation testing failure as a result of insufficient quantity or quality of sample, or technical failure occurred at sequencing or postsequencing steps on analysis.

A key secondary end point was overall survival. Other secondary end points reported included time to first subsequent therapy or death (TFST), time to second progression or death (PFS2), and health-related quality of life (HRQoL) using the Functional Assessment of Cancer Therapy-Prostate Cancer (FACT-P) questionnaire. FACT-P scores range from 0 to 156, with higher scores indicating better HRQoL for men with prostate cancer (full definitions are provided in the Supplementary Appendix, p. 9).

Exploratory end points included objective response rate (ORR), prostate-specific antigen (PSA) response rate, and time to PSA progression (full list of end points in Supplementary Appendix, p. 8).

Safety was assessed by reporting adverse events and serious adverse events (Common Terminology Criteria for Adverse Events version 4.03) based on physical examination findings, vital signs, electrocardiogram findings, and laboratory test results.

TRIAL OVERSIGHT

This trial was performed in accordance with the principles of the Declaration of Helsinki, Good Clinical Practice guidelines, and the AstraZeneca and Merck Sharp &

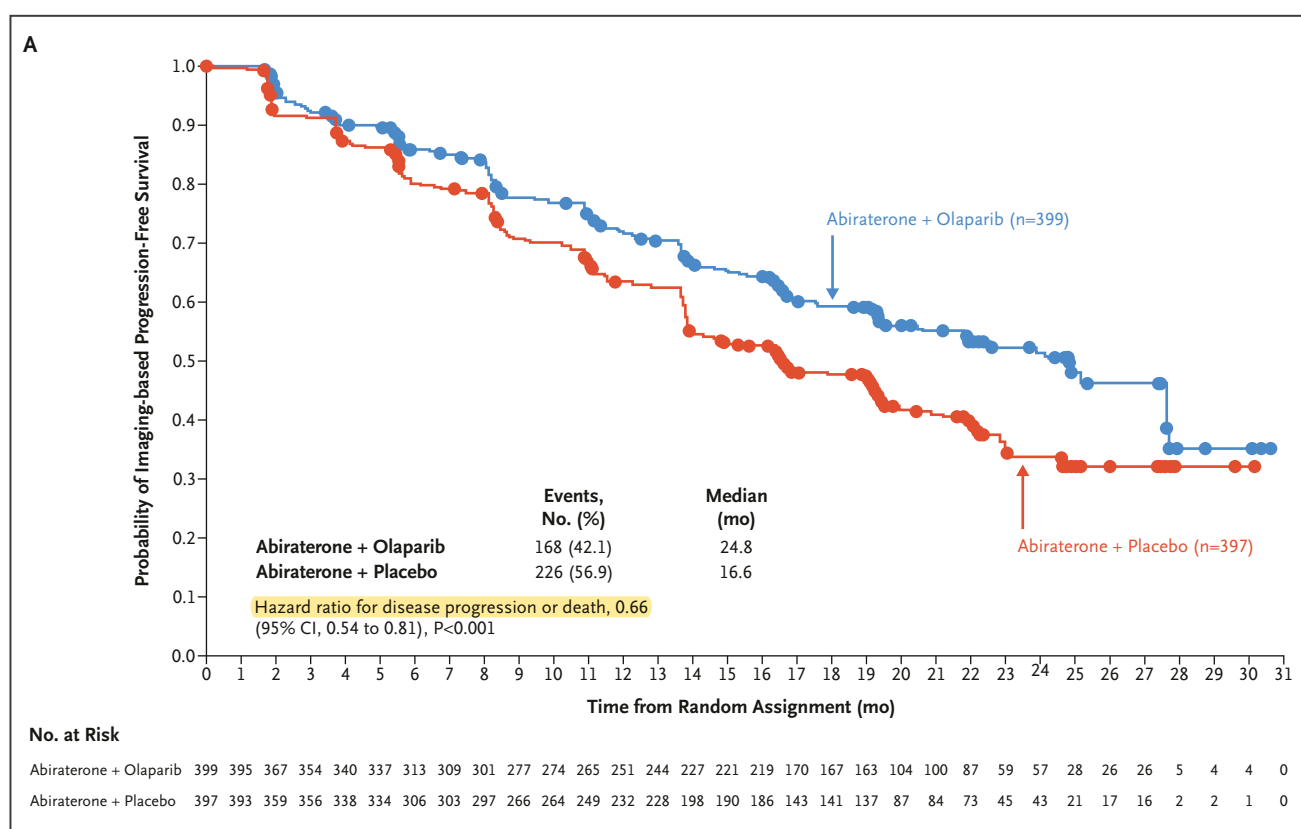


Figure 1. Kaplan-Meier Estimates of Imaging-Based Progression-Free Survival.

Panel A shows imaging-based progression-free survival by investigator assessment. There were 168 progression or death events in the abiraterone and olaparib arm and 226 in the abiraterone and placebo arm. The median (range) duration of follow-up for disease progression in patients with censored data was 19.3 months (0.03 to 30.59 months) in the abiraterone and olaparib arm and 19.4 months (0.03 to 30.16 months) in the abiraterone and placebo arm. Panel B shows imaging-based progression-free survival by blinded independent central review. There were 157 progression or death events in the abiraterone and olaparib arm and 218 in the abiraterone and placebo arm. The median (range) duration of follow-up for disease progression in patients with censored data was 19.3 months (0.03 to 30.59 months) in the abiraterone and olaparib arm and 19.2 months (0.03 to 30.16 months) in the abiraterone and placebo arm. A circle indicates a censored observation. CI denotes confidence interval.

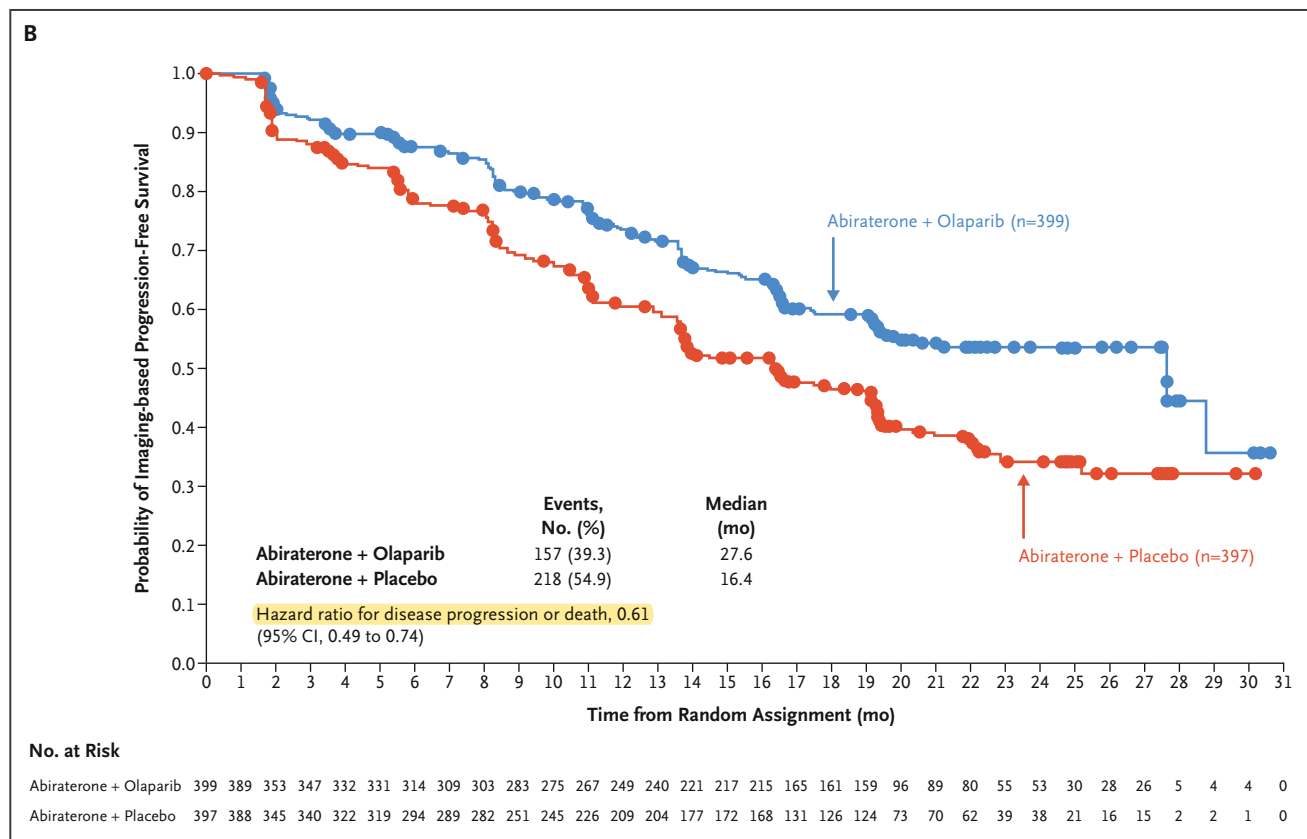


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Dohme, LLC, policies on bioethics. It was designed by representatives of AstraZeneca and Merck Sharp & Dohme, LLC, with consultation from the trial standing advisory board. AstraZeneca was responsible for overseeing the collection, analysis, and interpretation of the data. All authors had full data access. The manuscript was written with medical writing assistance funded by AstraZeneca and Merck Sharp & Dohme, LLC, with critical review by and input from the authors. The authors attest to the accuracy and completeness of the data and the fidelity of the trial to the protocol.

STATISTICAL ANALYSIS

Efficacy was analyzed for the intention-to-treat population, and safety was analyzed for all patients who received any amount of abiraterone, olaparib, or placebo. Patients who received at least one dose of olaparib were included in the abiraterone and olaparib arm.

With a sample size of 796 patients, the primary analysis at a first data cutoff was planned to occur when there had been approximately 379 progression or death events

(47.6% maturity), to provide 94.1% power at a one-sided alpha of 0.014 to show a statistically significant difference in ibPFS between the trial arms, assuming a hazard ratio for progression or death of 0.68.

A multiplicity testing procedure based on the graphical approach in group sequential trials of Maurer and Bretz,¹⁸ analogous to a simple sequential gatekeeping method, strongly controlled the overall familywise one-sided error rate of 2.5%. If the primary end point of ibPFS was statistically significant, then overall survival would be tested in a hierarchical fashion (Fig. S1 in the Supplementary Appendix).

The O'Brien and Fleming spending function,¹⁹ calculated on the basis of actual observed events, was used to control the overall type I error, with the restriction that alpha spend for the overall survival interim analysis at the first data cutoff would not exceed 0.0005.

For time-to-event end points, a stratified log-rank test was used to calculate two-sided P values. Hazard ratios and 95%

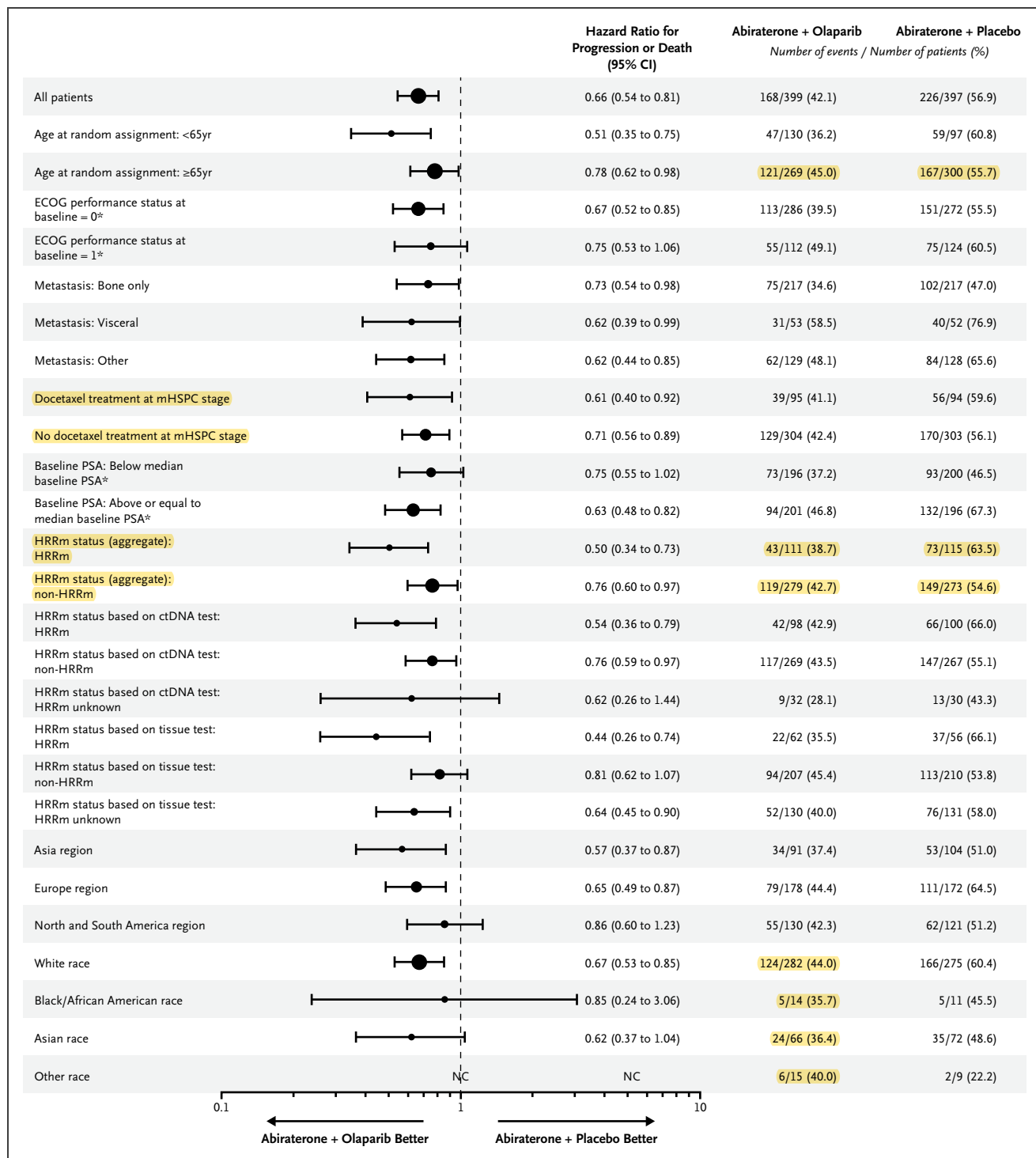


Figure 2. Forest Plot of Prespecified Subgroup Analysis of Imaging-Based Progression-Free Survival (Investigator Assessed).

Data were derived from Interactive Voice/Web Response System stratification variables. Analysis performed included the stratification factors selected in the primary pooling strategy as covariates. Each subgroup analysis was performed using a Cox proportional hazards model that contained a term for treatment, factor, and treatment by factor interaction. A hazard ratio <1 implies a lower risk of progression on olaparib. The size of a circle is proportional to the number of events. Subgroup categories with fewer than five events in either treatment group have NC presented. *Excludes patients with no baseline assessment. CI denotes confidence interval, ctDNA circulating tumor DNA, ECOG Eastern Cooperative Oncology Group, HRRm homologous recombination repair gene mutation, mHSPC metastatic hormone-sensitive prostate cancer, NC noncalculable, and PSA prostate-specific antigen.

CI's were calculated using the Cox proportional hazards model including the two stratification variables as covariates. Kaplan-Meier plots were used to calculate medians. The statistical analysis plan is available with the trial protocol.

Results

SCREENING AND RANDOM ASSIGNMENT

This trial spanning 17 countries screened 1103 patients between October 2018 and January 2020; 796 patients met the eligibility criteria and were randomly assigned to study treatment. For details on screening, random assignment, and follow-up, see Figure S2.

PATIENT CHARACTERISTICS

Overall, 399 patients were assigned to abiraterone and olaparib and 397 to abiraterone and placebo. Baseline

characteristics were generally well balanced between arms (Table 1 and Table S1). On the basis of aggregate HRRm testing (tumor tissue and ctDNA), 28.4% and 69.3% of patients were included in the HRRm and non-HRRm subgroups, respectively; 2.3% of patients had unknown HRRm status. The proportions of patients with *BRCA1* and *BRCA2* mutations in the trial arms were consistent with previously reported prevalence in mCRPC^{17,20} and were well balanced between arms. The representatives of the enrolled patients are described in Table S2.

EFFICACY

Primary End Point

This primary analysis was undertaken at the prespecified first data cutoff after 394 patients had an imaging-based progression event or had died (49.5% maturity, data cutoff of July 30, 2021). The median ibPFS by investigator assessment was significantly longer for the abiraterone and olaparib arm than

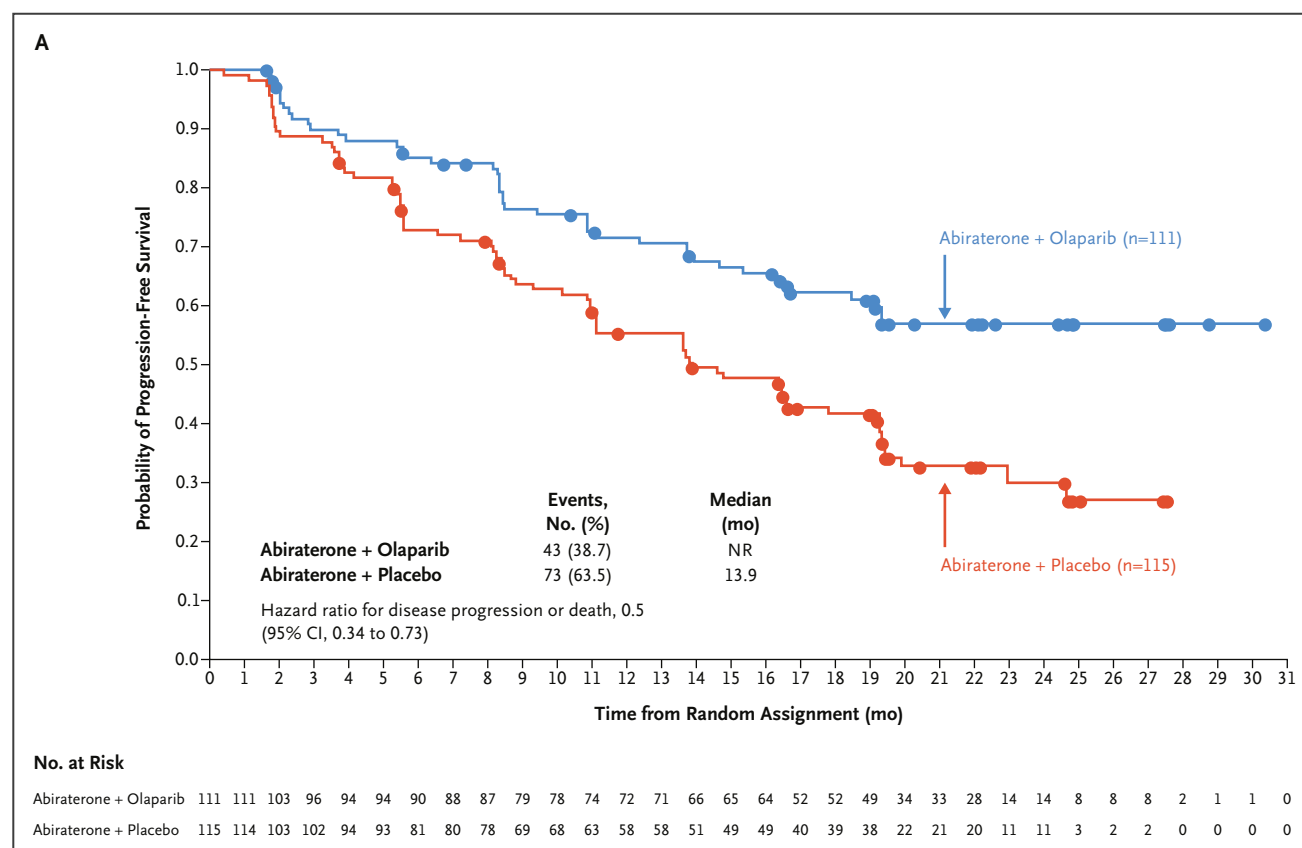


Figure 3. Kaplan-Meier Estimates of HRRm Subgroup Analysis of Imaging-Based Progression-Free Survival (Aggregate Tumor Tissue and Circulating Tumor DNA).

Homologous recombination repair gene mutation (HRRm) subgroup by investigator assessment (Panel A) or by blinded independent central review (Panel B) is shown. Non-HRRm subgroup by investigator assessment (Panel C) or by blinded independent central review (Panel D) is shown. A circle indicates a censored observation. CI denotes confidence interval and NR denotes not reached.

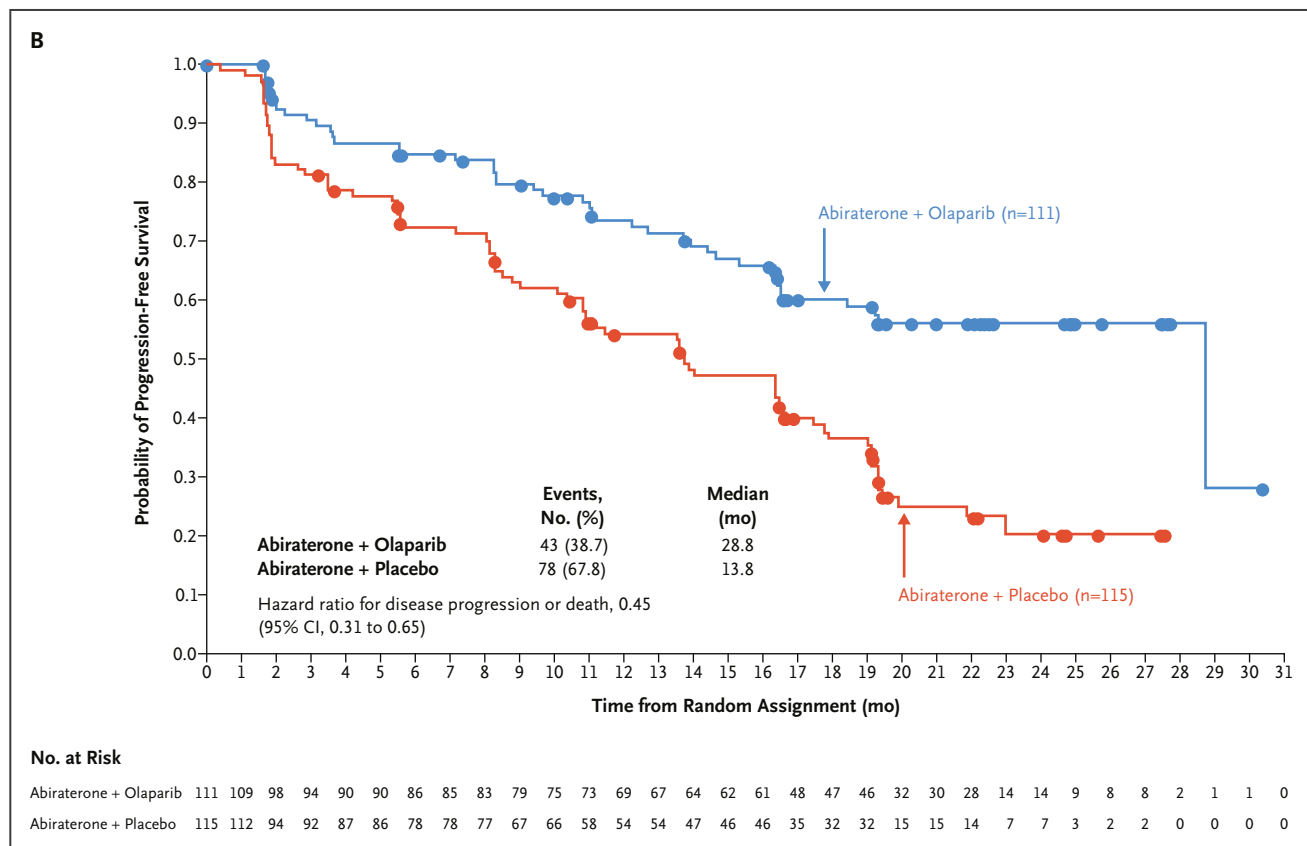


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for the abiraterone and placebo arm (24.8 vs. 16.6 months; hazard ratio, 0.66; 95% CI, 0.54 to 0.81; $P < 0.001$) (Fig. 1A).

Prespecified sensitivity analysis of ibPFS by blinded independent central review was consistent with the results of the primary analysis (median, 27.6 vs. 16.4 months; hazard ratio, 0.61; 95% CI, 0.49 to 0.74) (Fig. 1B).

Exploratory Subgroup Analysis of the Primary End Point

An ibPFS benefit was observed across all prespecified subgroups whether evaluated by investigator assessment or by blinded independent central review (Fig. 2 and Fig. S3). A global interaction test comparing the fit of a model with no interaction terms with a model with all subgroup interactions included was not significant at the 10% level ($P = 0.41$), indicating a consistent treatment effect between subgroups.

HRRm status was established for 535 patients (67.2%) by tumor tissue test, 734 (92.2%) by ctDNA test, and 778

(97.7%) by aggregated tumor tissue and ctDNA test results. The aggregate HRRm population included 226 patients (90 positive by tumor tissue and ctDNA, 28 positive by tumor tissue, and 108 positive by ctDNA), and the non-HRRm population included 552 patients (328 negative by tumor tissue and ctDNA, 38 negative by tumor tissue, and 186 negative by ctDNA).

All hazard ratios for the HRRm and non-HRRm populations favored the combination of abiraterone and olaparib versus abiraterone and placebo (e.g., HRRm [aggregate tumor tissue and ctDNA] subgroup, ibPFS by investigator assessment: hazard ratio, 0.50; 95% CI, 0.34 to 0.73; and non-HRRm: hazard ratio, 0.76; 95% CI, 0.60 to 0.97; Fig 2; and blinded independent central review in Fig. S3). Median ibPFS in both the HRRm and non-HRRm populations also suggested an improvement of ibPFS in these populations (Fig. 3).

Secondary End Points

Overall survival data were immature at this primary analysis at the first data cutoff (28.6% maturity; hazard ratio, 0.86;

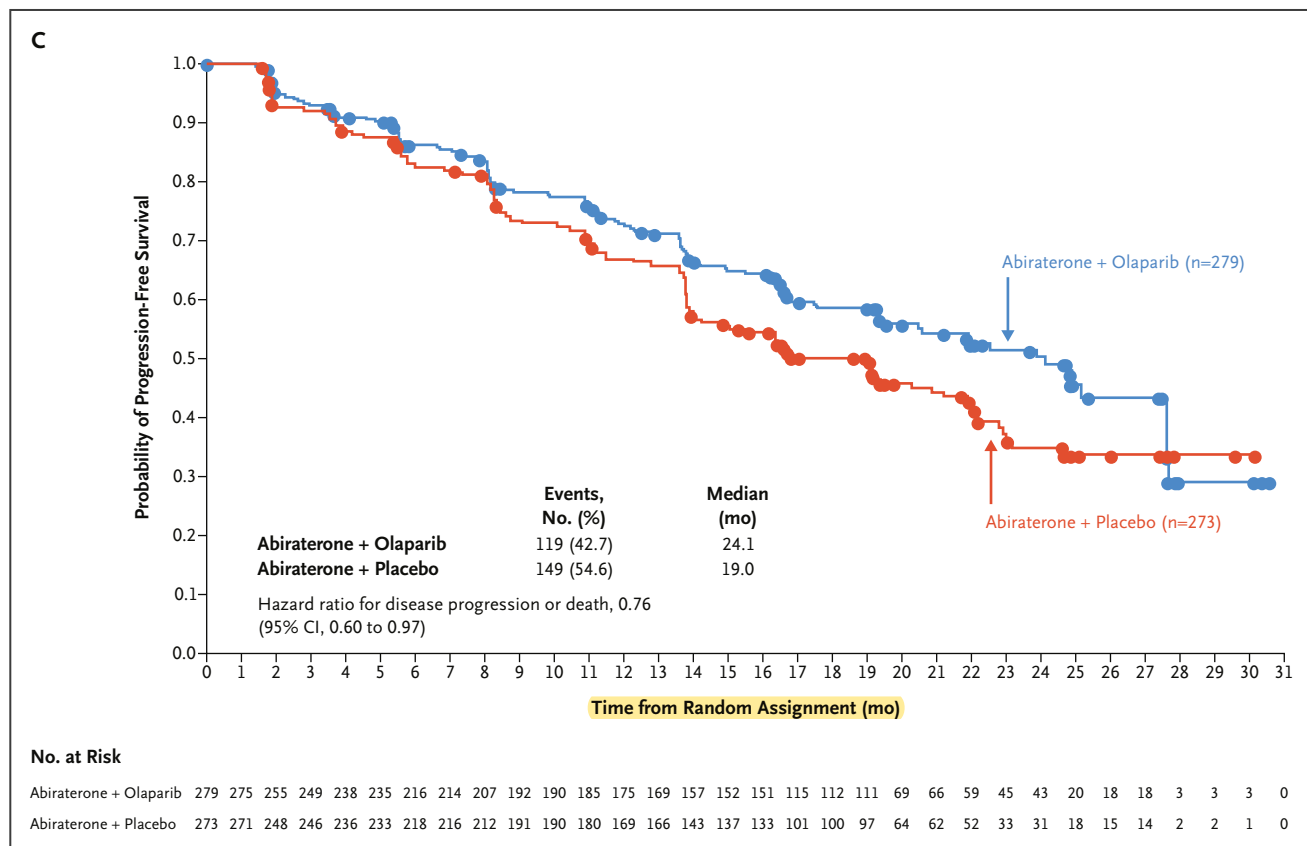


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95% CI, 0.66 to 1.12; $P=0.29$) (Fig. 4A). However, TFST (hazard ratio, 0.74; 95% CI, 0.61 to 0.90) and PFS2 (hazard ratio, 0.69; 95% CI, 0.51 to 0.94) were supportive of an efficacy beyond first imaging-based progression (Fig. 4B and 4C). Overall, 305 patients received subsequent therapies (132 in the abiraterone and olaparib arm and 173 in the abiraterone and placebo arm), of which approximately two thirds were docetaxel or cabazitaxel and one third were next-generation hormonal agents (Table S3).

Least-square mean change from baseline in FACT-P total score across all visits was -4.85 in the abiraterone and olaparib arm versus -4.03 in the abiraterone and placebo arm (difference, -0.82 ; 95% CI, -3.56 to 1.92).

Exploratory End Points

Of 40.3% of patients with measurable disease at baseline, the ORR was 58.4% (94 of 161 patients) in the abiraterone and olaparib arm versus 48.1% (77 of 160 patients) in the abiraterone and placebo arm (odds ratio, 1.60; 95% CI, 1.02 to 2.53; Table S4).

Confirmed PSA response was 79.3% in the abiraterone and olaparib arm and 69.2% in the abiraterone and placebo arm. Median time to PSA progression was not reached with abiraterone and olaparib versus 12.0 months with abiraterone and placebo (hazard ratio, 0.55; 95% CI, 0.45 to 0.68).

Safety

At the data cutoff, median total duration of exposure was 17.5 months for olaparib, 15.7 months for placebo, 18.2 months for abiraterone in the abiraterone and olaparib arm, and 15.7 months for abiraterone in the abiraterone and placebo arm.

The most common adverse events in the abiraterone and olaparib arm were anemia, fatigue/asthenia, and nausea. Anemia was the most common grade 3 or higher adverse event, occurring in 60 patients (15.1%) in the abiraterone and olaparib arm and 13 patients (3.3%) in the abiraterone and placebo arm.

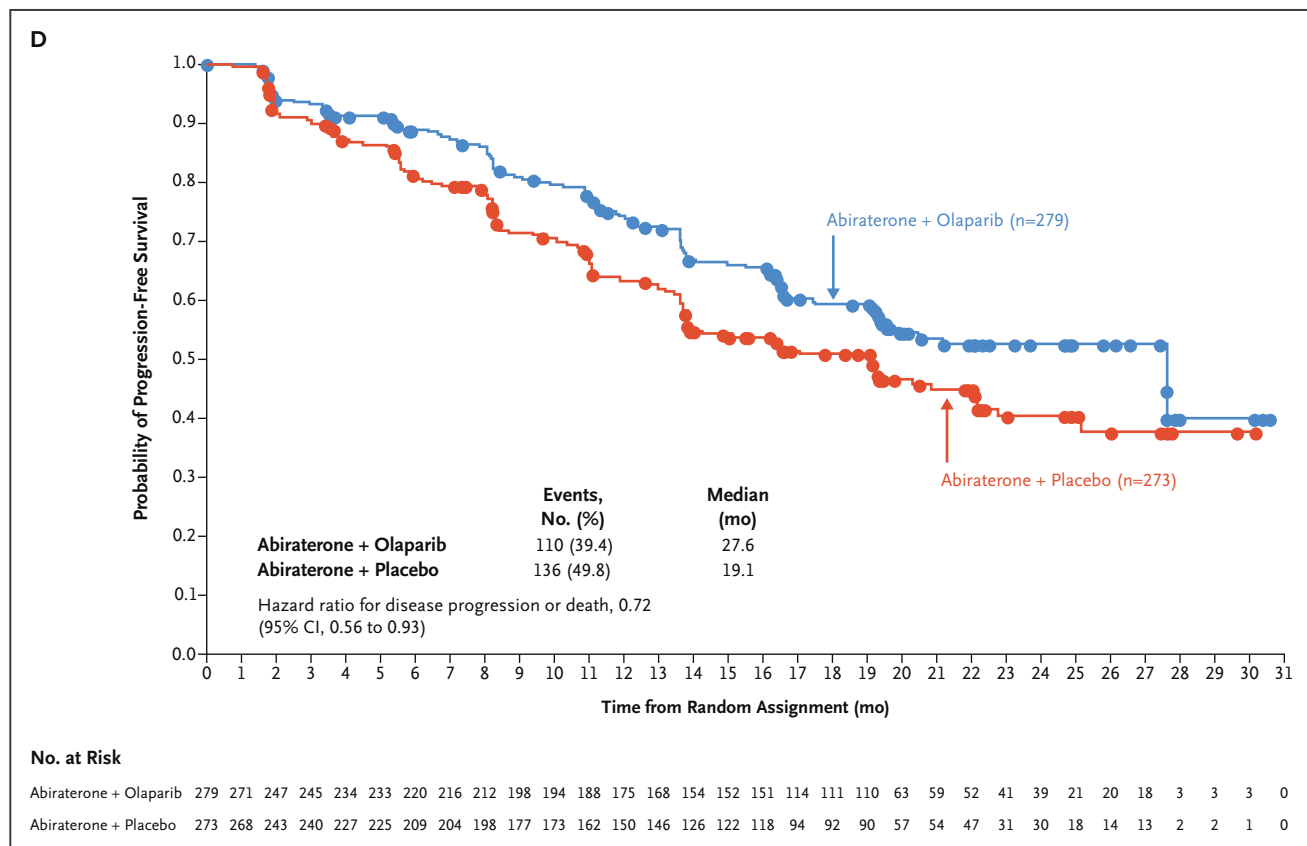


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Fifty-five patients (13.8%) discontinued olaparib and 31 patients (7.8%) discontinued placebo because of an adverse event. Discontinuations of abiraterone as a result of adverse events occurred in 34 patients (8.5%) and 35 patients (8.8%) in the abiraterone and olaparib and abiraterone and placebo arms, respectively.

The rate of cardiovascular events (myocardial infarction, congestive heart failure, and ischemic stroke) was similar between the treatment arms (Table 2 and Tables S6 and S7).

Twenty-six cases of pulmonary embolism occurred (6.5% of patients) in the abiraterone and olaparib arm and seven (1.8% of patients) in the abiraterone and placebo arm; one event in the abiraterone and olaparib arm was fatal. Whether this was attributable to the play of chance or to a true difference in the incidence of venous thromboembolic events is not known. However, the overall incidence of pulmonary embolism observed in the PROpel trial is consistent with the incidence reported in patients with prostate cancer in the literature.^{7,21,27} Pulmonary embolism led

to dose interruption of olaparib in eight patients (2.0%) and abiraterone in six patients (1.5%) in the abiraterone and olaparib arm; these events did not lead to treatment discontinuation (Table S8; Supplementary Appendix, p. 20). Deep-vein thrombosis occurred in seven patients (1.8%) and three patients (0.8%) in the abiraterone and olaparib and abiraterone and placebo arms, respectively.

No patients developed myelodysplastic syndrome or acute myeloid leukemia (p. 21 of the Supplementary Appendix includes a summary of bone marrow or blood cytogenetic analysis). There were 12 reports (3.0% of patients) of new primary cancers in the abiraterone and olaparib arm and 10 reports (2.5% of patients) in the abiraterone and placebo arm. Three patients (0.8%) in each arm of the trial had new-onset pneumonitis (one in each arm had interstitial lung disease).

There were 33 cases (8.3% of patients) of Covid-19 in the abiraterone and olaparib arm and 18 cases (4.5% of patients) in the abiraterone and placebo arm. Of patients who developed Covid-19, none were fully vaccinated by the case definition for each vaccine product used before

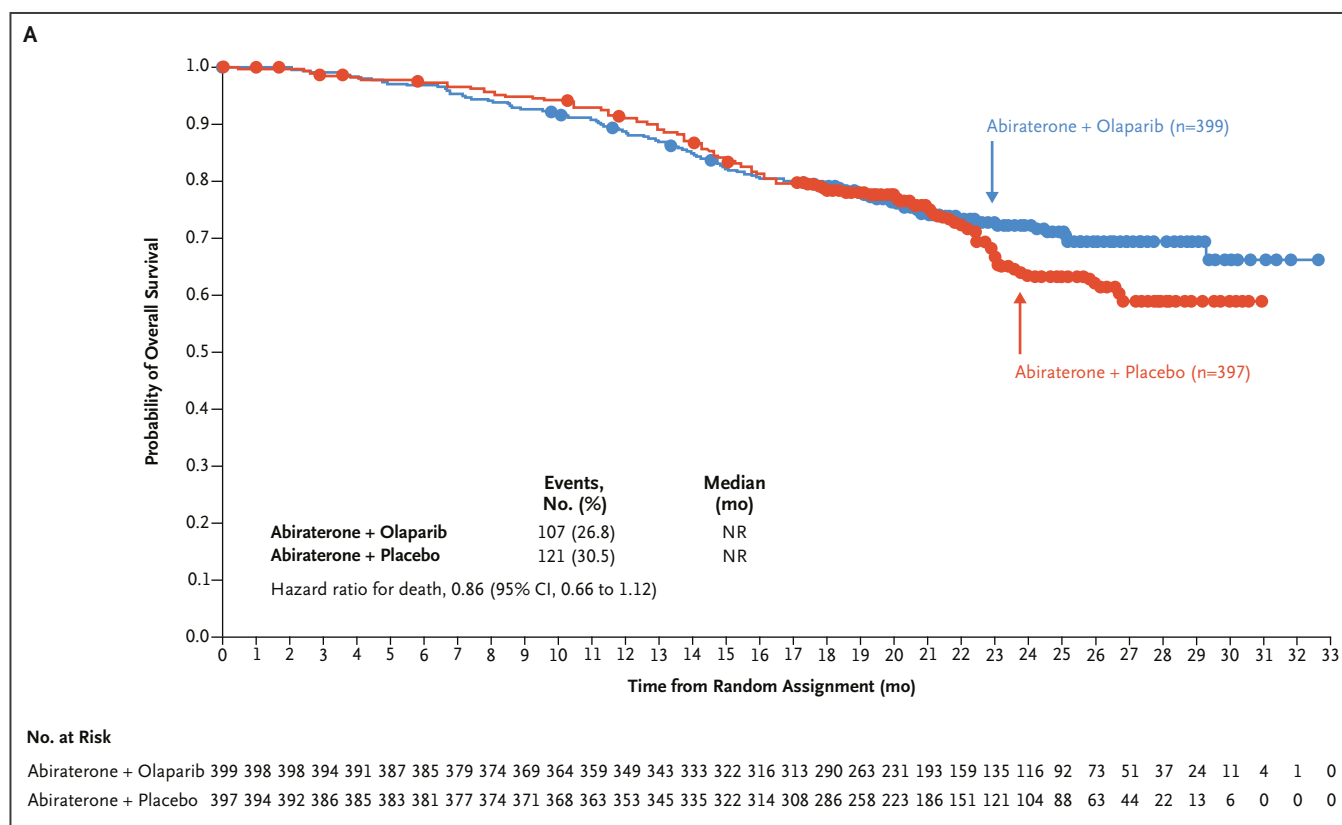


Figure 4. Kaplan-Meier Estimates of the Secondary End Points of Overall Survival, Time to First Subsequent Therapy and Time to Second Progression.

Panel A shows overall survival by investigator assessment. There were 107 deaths in the abiraterone and olaparib arm and 121 in the abiraterone and placebo arm. The median (range) duration of follow-up for disease progression in patients with censored data was 22.2 months (0.03 to 32.56 months) in the abiraterone and olaparib arm and 21.8 months (0.10 to 30.88 months) in the abiraterone and placebo arm. Panel B shows time to first subsequent therapy or death. There were 183 patients who received a first subsequent therapy in the abiraterone and olaparib arm and 221 patients in the abiraterone and placebo arm. Time to first subsequent therapy (excluding radiotherapy) was defined as the time from random assignment to the start date of the first subsequent anticancer therapy after discontinuation of randomized treatment or death from any cause (whichever was earlier). Any patient not known to have died at the time of the analysis and not known to have had a subsequent therapy was censored at the last known time to have not received first subsequent therapy. Panel C shows time to second progression or death by investigator assessment. There were 70 second progression or death events in the abiraterone and olaparib arm and 94 in the abiraterone and placebo arm. Patients who had not had a second disease progression event or had died at the time of analysis, or who had second progression or had died after two or more missed visits, were censored at the latest evaluable assessment when they were known to be alive and without a second disease progression. PFS2 was defined as the time from random assignment to second progression on next-line anticancer therapy by investigator assessment of imaging-based progression, or clinical symptomatic progression. A circle indicates a censored observation. CI denotes confidence interval and NR not reached.

being diagnosed with Covid-19. Within the abiraterone and olaparib arm, 13 patients were from Brazil, 4 were from Turkey, and 5 were from the United States.

Discussion

At the primary analysis at this first data cutoff, the PROpel trial met its primary end point. In patients receiving

first-line treatment for mCRPC, enrolled irrespective of HRRm, treatment with abiraterone and olaparib significantly prolonged ibPFS compared with abiraterone and placebo.

The median ibPFS (24.8 vs. 16.6 months by investigator assessment and 27.6 vs. 16.4 months by blinded independent central review in the abiraterone and olaparib and abiraterone and placebo arms, respectively) is a substantial

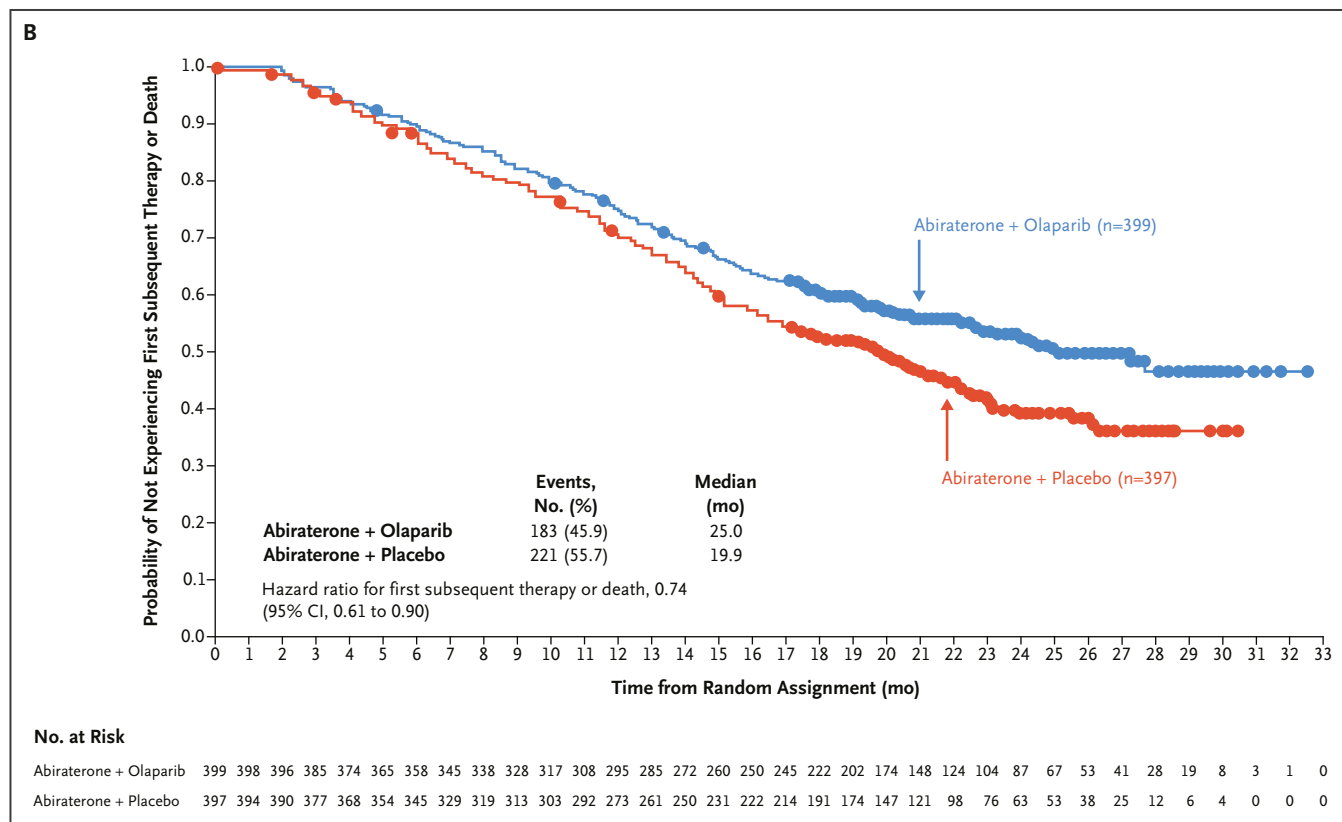


Figure 4. Continued.

improvement of this outcome in this population including patients with symptomatic, asymptomatic, or mildly symptomatic disease and those with visceral metastases (provided they were considered candidates for abiraterone by the investigator).

Because enrollment was not based on biomarker status, patients were representative of those eligible for abiraterone in the real-world setting without restricting therapy to patients for whom HRRm status could be assigned. The active control arm of abiraterone plus prednisone/prednisolone and placebo performed as expected. In the COU-AA-302 study in patients with progressive mCRPC who had not received chemotherapy, the median ibPFS with abiraterone was 16.5 months.²² This compares with the 16.6 months reported herein, illustrating that combining abiraterone and olaparib treatment significantly extended ibPFS beyond the current standard of care.

We believe that the changes we observed represent clinically meaningful improvement in ibPFS in all prespecified

subgroups, including subgroups for prior and no prior treatment with docetaxel at the mHSPC stage of disease, metastasis type at baseline, and HRRm and non-HRRm subgroups (whether these were evaluated by tumor tissue, ctDNA, or aggregate). Notably, the tumor tissue test failure rate observed in PROpel (31.6%) was in line with that observed in PROfound (31%)⁶ and other published literature.¹⁷ However, the concordance between tumor tissue and ctDNA tests, as well as analysis of the aggregate tumor tissue and ctDNA data set, gives confidence to the assignment of patients to the non-HRRm subgroup. On the basis of our calculations, approximately 7 to 13 of 552 non-HRRm cases in the aggregate analysis could be false negatives resulting from the recognized performance limitations of plasma ctDNA testing (Supplementary Appendix, p. 21).

Our findings validate the radiographic progression-free survival results of the earlier phase 2 trial of abiraterone and olaparib versus abiraterone and placebo in patients with mCRPC who had previously received docetaxel (hazard ratio, 0.65; 95% CI, 0.44 to 0.97; $P=0.034$).¹¹ They

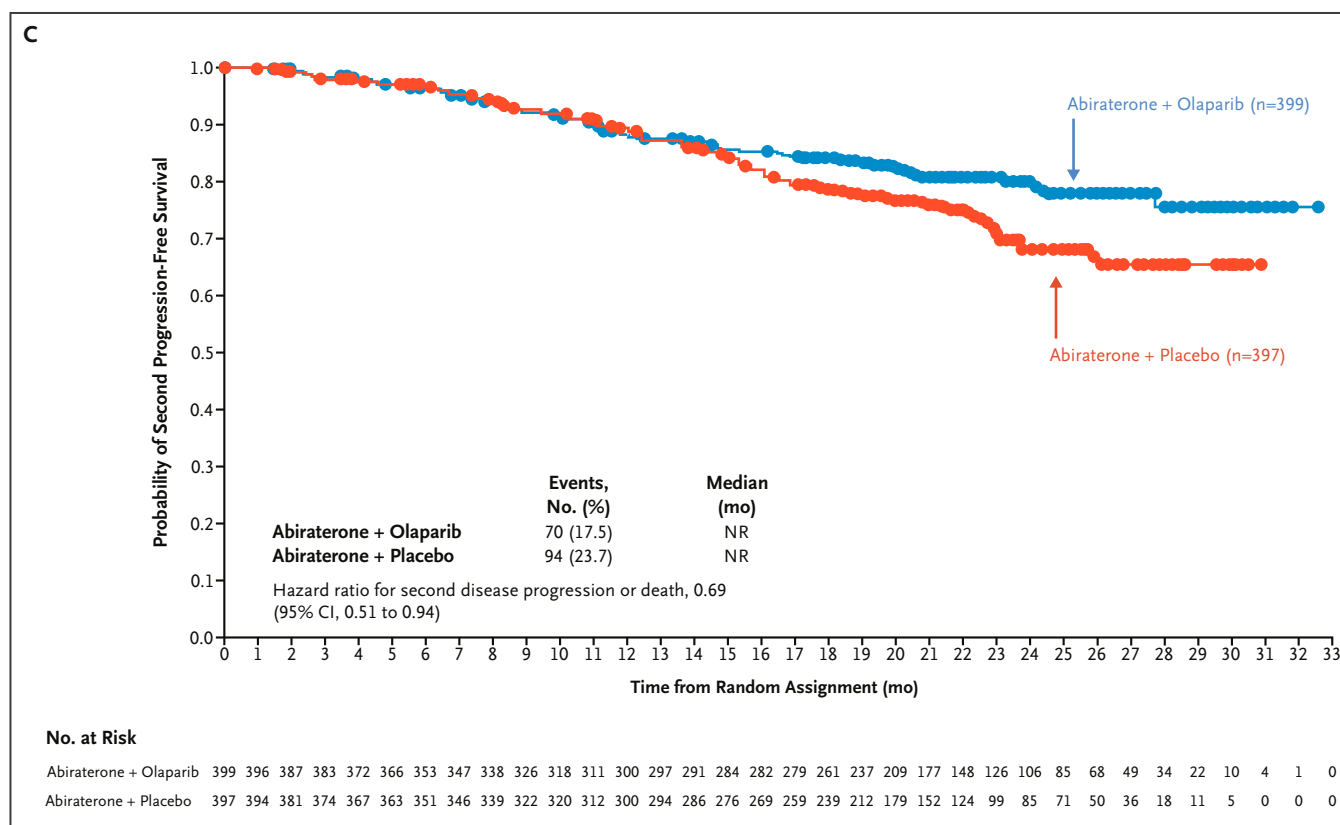


Figure 4. Continued.

are also consistent with preclinical studies that demonstrated a combined antitumor effect with next-generation hormonal agents and PARP inhibition.^{8,9}

A phase 2 trial of the PARP inhibitor veliparib in combination with abiraterone versus abiraterone found no significant difference in efficacy outcomes for patients with mCRPC when veliparib was added to abiraterone.²³ The different findings of the phase 2 veliparib trial and the PROpel and the previous phase 2 abiraterone and olaparib trial may be explained by the weaker PARP-trapping activity of veliparib compared with olaparib observed in preclinical models and by their differing abilities to interact with other PARP family members.^{11,24,25} Dosing may also play a role in the activity of abiraterone and olaparib in the non-HRRm subgroup; PROpel patients were able to receive full monotherapy doses of abiraterone and olaparib because these drugs do not have clinically relevant drug-drug interactions¹¹ or major overlapping toxicities.^{2,7}

A retrospective analysis of 15 studies demonstrated a strong correlation between PFS2 and overall survival,

supporting the use of PFS2 to measure long-term clinical benefit when overall survival cannot be assessed.²⁶ Although overall survival data and PFS2 data were immature at this data cutoff, Kaplan-Meier estimates for both end points showed a similar trend favoring the abiraterone and olaparib combination. In alignment with the course of disease progression in the clinical setting, the separation of the treatment arms in the Kaplan-Meier estimates occurred earlier for PFS2 than for overall survival. The Kaplan-Meier estimate for TFST (50.8% maturity) showed clear separation of the treatment arms and suggested a delay of 5.1 months to the next subsequent therapy with the combination treatment, highlighting clinical benefit beyond ibPFS.

There were more adverse events in the abiraterone and olaparib trial arm, particularly anemia. However, the adverse-event profile for abiraterone and olaparib was consistent with their known individual toxicity profiles and did not suggest that combination therapy increased the toxicity of either drug. In the phase 2 trial of abiraterone and olaparib versus abiraterone and placebo in

Table 2. Treatment-Emergent Adverse Events (Safety Analysis Set).*

Adverse Event	Abiraterone and Olaparib (n=398)		Abiraterone and Placebo (n=396)	
	All Grades	Grade ≥3	All Grades	Grade ≥3
Any	387 (97.2)	188 (47.2)	376 (94.9)	152 (38.4)
Any serious	135 (33.9)	NA	107 (27.0)	NA
Interruption due to adverse event				
Olaparib or placebo	178 (44.7)	NA	100 (25.3)	NA
Abiraterone	131 (32.9)	NA	87 (22.0)	NA
Dose reduction due to adverse event				
Olaparib or placebo	80 (20.1)	NA	22 (5.6)	NA
Abiraterone	10 (2.5)	NA	17 (4.3)	NA
Discontinuation due to adverse event				
Olaparib or placebo	55 (13.8)	NA	31 (7.8)	NA
Abiraterone	34 (8.5)	NA	35 (8.8)	NA
Death due to adverse event	16 (4.0)	NA	17 (4.3)	NA
Type (in ≥10% of patients in abiraterone and olaparib arm)				
Anemia†	183 (46.0)	60 (15.1)	65 (16.4)	13 (3.3)
Fatigue or asthenia	148 (37.2)	9 (2.3)	112 (28.3)	6 (1.5)
Nausea	112 (28.1)	1 (0.3)	50 (12.6)	1 (0.3)
Diarrhea	69 (17.3)	3 (0.8)	37 (9.3)	1 (0.3)
Constipation	69 (17.3)	0	55 (13.9)	1 (0.3)
Back pain	68 (17.1)	3 (0.8)	73 (18.4)	4 (1.0)
Decreased appetite	58 (14.6)	4 (1.0)	23 (5.8)	0
Vomiting	52 (13.1)	4 (1.0)	36 (9.1)	1 (0.3)
Arthralgia	51 (12.8)	0	70 (17.7)	2 (0.5)
Hypertension	50 (12.6)	14 (3.5)	65 (16.4)	13 (3.3)
Dizziness	43 (10.8)	0	25 (6.3)	0
Peripheral edema	41 (10.3)	0	45 (11.4)	1 (0.3)
Urinary tract infection	41 (10.3)	8 (2.0)	31 (7.8)	4 (1.0)
Other				
Cardiac failure‡	6 (1.5)	4 (1.0)	5 (1.3)	1 (0.3)
Embolic and thrombotic events				
Arterial‡	8 (2.0)	6 (1.5)	10 (2.5)	8 (2.0)
Venous‡	29 (7.3)	27 (6.8)	13 (3.3)	8 (2.0)

* Patients were counted once for each type of adverse event and are reported regardless of the investigators' assessment of causality. Adverse events with an onset date, or worsening, on or after the date of first dose and up to and including 30 days after discontinuation of randomized treatment were included. Common Terminology Criteria for Adverse Events version 4.03 was used. NA denotes not applicable.

† Anemia included anemia, decreased hemoglobin level, decreased red blood cell count, decreased hematocrit level, erythropenia, macrocytic anemia, normochromic anemia, normochromic normocytic anemia, and normocytic anemia.

‡ Based on standardized MedDRA query.

mCRPC, more patients had cardiovascular events in the abiraterone and olaparib arm. In PROpel, a difference in cardiovascular events was not observed between the two arms, suggesting that the imbalance in the phase 2 trial may have been due to its small population size. In the PROpel trial, there was also an imbalance in pulmonary events between trial arms, and one event in the abiraterone and olaparib arm was fatal. Pulmonary embolism events have been observed in PARP inhibitor monotherapy

trials in mCRPC. In the PROfound trial, 5% of patients in the olaparib arm and 0.8% of patients in the control arm of abiraterone or enzalutamide had a pulmonary embolism; in the TALAPRO-1 trial, 6% of patients had a pulmonary embolism.^{7,27} Currently, the mechanism of this effect is unknown. There were more cases of Covid-19 in the olaparib and abiraterone arm of trial; the reasons for this imbalance are too complex to ascertain because of differences in the evolution of the pandemic across

countries and in the responses and restrictions imposed regionally.

We describe the primary analysis at first data cutoff in which abiraterone and olaparib led to significantly longer ibPFS than did abiraterone and placebo in patients with mCRPC, enrolled irrespective of HRRm status, who had not received treatment in the first-line setting. Overall survival at this data cutoff did not reach the prespecified threshold for significance but is only 28.6% mature. Moreover, TFST, PFS2, ORR, and PSA response support the treatment benefit of abiraterone and olaparib over abiraterone and placebo (Table S5). There were more adverse events, particularly anemia and pulmonary embolism, in the abiraterone and olaparib arm, although these were as expected for the individual drugs, and there was no clinically meaningful detriment to HRQoL. The overall results of PROpel demonstrate the clinical benefits of olaparib in combination with abiraterone in the first-line treatment of a broad, HRRm unselected population of patients with mCRPC.

Disclosures

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